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Factoring Out the Greatest Common Factor

When finding the factors of a number, you look for values that multiply to make that number. For example, 3 and 5 are factors of 15 because $3 \times 5 = 15$.

Factoring a polynomial expression is the opposite of using the distributive property. When factoring a polynomial, you look for the greatest common factor (GCF) of all the terms in the expression. You then divide by, or factor out, the greatest common factor.

Example 1

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Factor the expression $6x^2 + 18x + 12$.

Step 1

Find the greatest common factor of the coefficients.

The GCF of 6, 18, and 12 is 6.

Step 2

Find the greatest common factor of the variables.

The third term has no variable, so there is no common variable factor.

Step 3

Divide each term by the GCF.

The GCF of the expression is 6.

$$\frac{6x^2}{6} + \frac{18x}{6} + \frac{12}{6} = x^2 + 3x + 2$$

Step 4

Write the factors as a product.

$$6(x^2 + 3x + 2)$$

Example 2

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Factor the expression $24x^4 + 48x^3 + 56x^2$.

Step 1

Find the greatest common factor of the coefficients.

The GCF of 24, 48, and 56 is 8.

Step 2

Find the greatest common factor of the variables.

What do x^4 , x^3 , and x^2 have in common? Each term has at least an x^2 .

Step 3

Divide each term by the GCF.

The GCF of the expression is $8x^2$.

$$\frac{24x^4}{8x^2} + \frac{48x^3}{8x^2} + \frac{56x^2}{8x^2} = 3x^2 + 6x + 7$$

Step 4

Write the factors as a product.

$$8x^2(3x^2 + 6x + 7)$$

You can always check your work by multiplying using the distributive property. You should get the original expression.

The greatest common factor of a polynomial expression is a monomial. This kind of factoring is called *factoring out a monomial* because you pull the greatest common factor to the outside of the expression.

Key Term

factoring

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Lesson 2: Factoring

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